

A Z -VISIBILITY REPRESENTATION OF $K_{6,6}$ MINUS A PERFECT MATCHING

ABSTRACT. We exhibit an integral-coordinate z -visibility representation of $K_{6,6}$ minus a perfect matching. This directly refutes Conjecture 6.5 of Bose et al. (1994), restated in the 1998 journal version of that report. A complete finite certificate is included.

1. COUNTEREXAMPLE

Bose et al. [2, Sec. 1.2] represent vertices by pairwise disjoint closed axis-parallel rectangles in horizontal planes. Two rectangles A and B are z -visible if there is a closed vertical cylinder of positive length and positive radius whose two end disks are contained in A and in B and whose intersection with every other rectangle of the arrangement is empty; the 1994 report words this as “a closed cylinder C of non-zero length and radius such that the ends of C are contained in R_i and R_j , the axis of C is parallel to the z axis, and the intersection of C with any other rectangle in the arrangement is empty” [1, Sec. 1]. Adjacency is required to hold if and only if the corresponding rectangles are z -visible, and a graph admitting such a representation is called *VRR-representable* there. Writing $K_{j,j} - M$ for $K_{j,j}$ minus a perfect matching, the report proves that $K_{5,5} - M$ is VRR-representable [1, Theorem 6.3] and then conjectures, in the same section, that

“ $K_{6,6} - M$ is not VRR-representable”

[1, Conjecture 6.5]; the conjecture was restated in [2, Sec. 5]. It carries no further qualifier there, in particular not the “directed version of” qualifier of the adjacent [1, Theorem 6.4].

Let

$$H = \{h_0, \dots, h_5\}, \quad V = \{v_0, \dots, v_5\},$$

and let G have edge set

$$E(G) = \{h_i v_j : 0 \leq i, j \leq 5, i \neq j\}.$$

Thus G is $K_{6,6}$ minus the perfect matching $\{h_i v_i : 0 \leq i \leq 5\}$.

Theorem 1. *The graph G has a z -visibility representation. Hence Conjecture 6.5 of Bose et al. [1, Conjecture 6.5] is false.*

2020 *Mathematics Subject Classification.* 05C62, 68U05.

Key words and phrases. z -visibility representation, visibility graph, complete bipartite graph, perfect matching, counterexample, three-dimensional visibility.

For each rectangle R , the table gives its height $z(R)$ and its projection $\pi(R)$ to the xy -plane; the rectangle itself is $\pi(R) \times \{z(R)\}$. The distinct heights make the rectangles pairwise disjoint.

R	$z(R)$	$\pi(R)$
h_0	0	$[6, 9] \times [0, 7]$
h_3	1	$[2, 5] \times [0, 7]$
v_4	2	$[4, 8] \times [1, 3]$
v_0	3	$[2, 4] \times [1, 6]$
v_5	4	$[4, 8] \times [3, 6]$
h_1	5	$[1, 8] \times [2, 5]$
v_3	6	$[2, 10] \times [2, 5]$
h_2	7	$[3, 7] \times [1, 6]$
v_1	8	$[3, 7] \times [1, 7]$
h_5	9	$[2, 8] \times [1, 3]$
h_4	10	$[2, 8] \times [3, 6]$
v_2	11	$[0, 8] \times [0, 4]$

2. VERIFICATION

For integers a, b , set

$$Q_{a,b} = (a, a+1) \times (b, b+1), \quad \mathcal{C}(R) = \{(a, b) \in \mathbb{Z}^2 : Q_{a,b} \subseteq \pi(R)\}.$$

For distinct rectangles A, B , let

$$\mathcal{I}(A, B) = \{R : \min\{z(A), z(B)\} < z(R) < \max\{z(A), z(B)\}\}$$

and

$$\mathcal{F}(A, B) = (\mathcal{C}(A) \cap \mathcal{C}(B)) \setminus \bigcup_{R \in \mathcal{I}(A, B)} \mathcal{C}(R).$$

Lemma 2 (Unit-cell criterion). *Let the rectangles of the arrangement lie at pairwise distinct heights, so that they are pairwise disjoint, and let every projection boundary lie on an integer grid line. Then two of them, A and B , are z -visible if and only if $\mathcal{F}(A, B) \neq \emptyset$.*

Proof. Let D be the horizontal base disk of a witnessing cylinder. Choose $p \in \text{int } D$ on no integer grid line, and let $Q_{a,b}$ be the open unit cell containing p . Every projection boundary lies on an integer grid line, so for every rectangle R , either $Q_{a,b} \subseteq \pi(R)$ or $Q_{a,b} \cap \pi(R) = \emptyset$. Since D lies in both endpoint projections and avoids every intermediate projection, $(a, b) \in \mathcal{F}(A, B)$.

Conversely, if $(a, b) \in \mathcal{F}(A, B)$, the closed disk of radius $1/4$ centered at $(a + 1/2, b + 1/2)$ lies in $Q_{a,b}$. Its vertical extrusion from A to B is a witnessing cylinder. $\square$

For each $i \neq j$, the following entry is an element of $\mathcal{F}(h_i, v_j)$; a dash marks the deleted matching edge.

	v_0	v_1	v_2	v_3	v_4	v_5
h_0	–	(6, 6)	(6, 0)	(8, 2)	(6, 1)	(6, 3)
h_1	(2, 2)	–	(1, 2)	(2, 2)	(4, 2)	(4, 3)
h_2	(3, 1)	(3, 1)	–	(3, 2)	(4, 1)	(4, 5)
h_3	(2, 1)	(3, 6)	(2, 0)	–	(4, 1)	(4, 3)
h_4	(2, 5)	(3, 3)	(2, 3)	(2, 3)	–	(7, 5)
h_5	(2, 1)	(3, 1)	(2, 1)	(2, 2)	(7, 1)	–

For every remaining pair A, B , the next table lists rectangles in $\mathcal{I}(A, B)$ whose cell sets collectively cover $\mathcal{C}(A) \cap \mathcal{C}(B)$. The symbol $\emptyset$ means that this intersection of cell sets is empty. Thus $\mathcal{F}(A, B) = \emptyset$ in every listed case.

pair	blockers	pair	blockers	pair	blockers
Pairs within H					
h_0, h_1	v_4, v_5	h_0, h_2	v_4, v_5	h_0, h_3	$\emptyset$
h_0, h_4	v_5	h_0, h_5	v_4	h_1, h_2	v_3
h_1, h_3	v_0, v_4, v_5	h_1, h_4	v_3	h_1, h_5	v_3
h_2, h_3	v_0, v_4, v_5	h_2, h_4	v_1	h_2, h_5	v_1
h_3, h_4	h_2, v_0	h_3, h_5	h_2, v_0	h_4, h_5	$\emptyset$
Pairs within V					
v_0, v_1	h_2	v_0, v_2	h_1, h_5	v_0, v_3	h_1
v_0, v_4	$\emptyset$	v_0, v_5	$\emptyset$	v_1, v_2	h_4, h_5
v_1, v_3	h_2	v_1, v_4	h_2	v_1, v_5	h_2
v_2, v_3	h_4, h_5	v_2, v_4	h_5	v_2, v_5	h_1
v_3, v_4	h_1	v_3, v_5	h_1	v_4, v_5	$\emptyset$
Deleted matching pairs					
h_0, v_0	$\emptyset$	h_1, v_1	h_2	h_2, v_2	v_1
h_3, v_3	h_1	h_4, v_4	$\emptyset$	h_5, v_5	$\emptyset$

Every entry of the two tables is verified by comparing integer coordinates. For the first table: the listed cell lies in $\mathcal{C}(h_i)$ and in $\mathcal{C}(v_j)$, and in $\mathcal{C}(R)$ for no $R \in \mathcal{I}(h_i, v_j)$. For the second: each listed blocker has height strictly between the two heights of its pair, and the listed blockers' cell sets together cover $\mathcal{C}(A) \cap \mathcal{C}(B)$, a cover that needs more than one blocker in nine of the rows; in the eight rows marked $\emptyset$ the set $\mathcal{C}(A) \cap \mathcal{C}(B)$ is itself empty, so no blocker is needed. By Lemma 2, the first table certifies all 30 edges of G , while the second excludes the other 36 pairs. Since

$$30 + 36 = \binom{12}{2},$$

the visibility graph of the twelve rectangles is exactly G . This proves Theorem 1.

Scope. What is settled is exactly [1, Conjecture 6.5]: it asserts that no representation of $K_{6,6} - M$ exists, and one representation refutes it. Nothing else is claimed. We prove nothing about $K_{j,j} - M$ for $j \geq 7$, and we make

no minimality claim: neither that twelve rectangles with these coordinate ranges are in any sense smallest, nor that the certificate above is the shortest possible.

The reading of the definition is load-bearing, and it is the source’s own: the end disks of the cylinder are *contained* in the two rectangles. Under the weaker reading in which those disks need only meet the two rectangles, these twelve rectangles realise a strictly larger graph and would not represent G ; so the containment reading is not a normalisation of convenience but the definition being refuted. The verification is, by contrast, insensitive to whether “meets another rectangle” is read as closed or as interior intersection, since each exhibited witness keeps clearance $1/4$ from every intermediate projection and an empty $\mathcal{F}(A, B)$ admits no cylinder of positive radius under either reading.

The heights above interleave the two sides, in the order $h_0, h_3, v_4, v_0, v_5, h_1, v_3, h_2, v_1, h_5, h_4, v_2$. The representation is therefore not a *directed* one in the sense of [1, Sec. 6], where representing a directed graph additionally requires the rectangle of each edge’s source to lie below the rectangle of its target, and nothing here bears on [1, Theorem 6.4], that the directed version of $K_{5,5} - M$ is not VRR-representable.

We have found no resolution of Conjecture 6.5 in the literature; it is restated as an open problem in [2, Sec. 5], the journal version of the report, published four years later. That is the outcome of a search, not a proof of priority. The 1994 report has no DOI and we found no library route to it; the copy read for this note, and the source of the wording quoted in Section 1, is the one at the address printed in [1]. A neighbouring model in the same literature, box visibility, where the channel may run in any of the three axis directions, is treated in the companion note [4], which exhibits nine unit cubes whose box-visibility graph is K_9 and so refutes Conjecture 26 of [3]. Different source, different statement, different model: neither result bears on the other.

Reproducibility. The inputs are exactly the data printed above: the twelve rectangles, the two certificate tables, the definitions of Section 2, and the combinatorial description of G . The counts 30, 36 and $66 = \binom{12}{2}$, the partition of the 66 pairs between the two tables, every cell set, and the visibility graph itself are recomputed rather than read off, and that the result is $K_{6,6}$ minus a perfect matching is confirmed without the labelling, from bipartiteness, 5-regularity, and complementarity in $K_{6,6}$ to a perfect matching. The visibility graph is built three ways—by the unit-cell criterion of Lemma 2, by the exact area of the free region over the arrangement of the printed coordinates, and by inclusion–exclusion over $\mathcal{I}(A, B)$ —and all three agree on the 66 pairs; for each of the 30 edges the closed disk of radius $1/4$ is checked in exact rational arithmetic to lie in both endpoint projections and to miss every intermediate one. The check discriminates: over the full census of all 96 admissible unit perturbations of a single projection

edge and all 66 transpositions of the height order, 79 and 61 respectively destroy the representation, and the three methods agree on all 162 mutants. All arithmetic is exact and nothing was sampled or reduced in range. The computation was re-run independently on a separate machine, where all 60 of its checks agreed; the program and its transcript are retained in an auxiliary archive available on request. Such agreement is evidence about the computation, not a second proof of Lemma 2, whose conclusion it confirms on this instance by the two methods that never use it.

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